Leaf-conditioned order in critical Tokunaga trees and strong Horton asymptotics for Kingman's coalescent

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Abstract

We give candidate proofs of refined asymptotics for two random-tree models. For every critical Tokunaga parameter $c > 1$, the expected Horton–Strahler order conditioned on n leaves is $\log_{2c} n + D_c(\log_{2c} n) + o(1)$, with an explicitly characterized continuous period-one function. More strongly, we obtain a phase-dependent conditional law in total variation and convergence of every fixed absolute moment. The argument uses an exact Galton–Watson-with-immigration representation and a weighted lattice local limit theorem. For Kingman's coalescent, the deterministic limiting branch densities satisfy both the ratio Horton law and $\mathcal{N}_j \rho^j \to H \in (0, \infty)$. The additional ingredients are monotone endpoint-normalized profiles and a fractional-linear comparison proving that $h_k(1) + 1$ is supermultiplicative. Numerical calculations are diagnostic only; no theorem depends on a floating-point output.

1 Scope and conventions

We address exact-leaf-conditioned order in critical Tokunaga trees and a positive geometric prefactor for Kingman's limiting branch densities: Problem 3 and the stronger-asymptotics part of Problem 9 in Kovchegov and Zaliapin's survey [6]. Leaves have Horton–Strahler order one. We use $R = 2c$ for the Tokunaga model and ρ for the Kingman constant, to avoid conflating two different constants.

For the Tokunaga model, the established inputs are the critical-process definition, independence of exponential edge lengths from tree shape, and the inverse-pruning construction (published survey Definition 25, p. 83; Proposition 11(2), p. 84; and Section 7.6.1, Lemma 12, p. 87). For Kingman, the input is the deterministic branch-density/ODE representation established in [5]. Fixed-order convergence of normalized finite-tree counts is already part of that result. The new issue is the subsequent limit in branch order, not a new fixed-order concentration theorem. We prove the needed analytic properties of the ODE hierarchy below, including the comparison used in the earlier root-Horton proof.

These arguments do not assert a joint limit for an arbitrary sequence $j = j(N)$ in finite Kingman trees, an almost-sure limit for every coupling of such trees, or a closed form for ρ.

1.1 Prior work and the precise contribution

The binary Catalan case is classical, including bounded periodic corrections in register-function asymptotics [1]. Here $c = 2$ is the critical binary Galton–Watson case, not a new model. With n leaves there are $n - 1$ internal vertices and $2n - 1$ total vertices; a statistic assigning zero to leaves is $K - 1$ in our convention. Thus conversion from an internal-vertex formula uses $n - 1$, not n, and conversion of an order-zero formula adds one. Since $\log_4(n - 1) - \log_4 n = o(1)$, these size conventions agree at the stated precision after the order shift. More explicitly, if that

convention gives $\mathbb{E}H = \log_4 m + F(\log_4 m) + o(1)$, then our correction is $D_2(\theta) = 1 + F(\theta)$, whenever F is continuous and period one. We do not claim originality for this specialization.

Brandenberger, Devroye and Reddad [2] establish first-order conditioned Galton–Watson growth. Khanfir [3, 4] extends this context to possibly infinite variance and develops fluctuations using a weighted real-valued statistic. Those results concern different model assumptions and observables. The present contribution is the integer-valued exact-leaf-conditioned local law and moment control for the full critical Tokunaga family $c > 1$, whose reduction below is to an auxiliary immigration chain, not an assertion that the trees are ordinary Galton–Watson trees.

For Kingman, the root-Horton exponent and endpoint comparison are established in [5]; the approximate numerical value of ρ is not new. Our additional steps are monotonicity of normalized profiles and a Möbius comparison ensuring a nonzero geometric prefactor. The literature comparison is bounded, not a guarantee of exhaustive priority. The two reusable mechanisms are weighted local bounds for exact conditioning and a structural normalization bound beyond ratio convergence.

I. Exact leaf conditioning

2 Critical Tokunaga trees: theorem and exact reduction

Let $K = \mathrm{ord}(T)$ and let L be the number of leaves in a critical Tokunaga tree. Its parameters are

$$\lambda_j = \gamma c^{1-j}, \qquad \mathbb{P}(K = k) = 2^{-k}, \qquad T_j = (c-1)c^{j-1}, \qquad c > 1, \quad \gamma > 0.$$

The scale γ does not affect the combinatorial distributions. Write X_k for the leaf count conditional on order k, and set $R = 2c$. At $c = 1$ the offspring mechanism is deterministic (perfect binary trees); the smoothing argument used here does not apply, so this boundary is excluded.

Theorem 2.1. *There is a strictly positive continuous density w_c on $(0, \infty)$ such that the series*

$$S_c(\theta) = \sum_{j \in \mathbb{Z}} (2R)^{-j} w_c(R^{\theta - j}), \tag{1}$$

$$D_c(\theta) = \frac{\sum_{j \in \mathbb{Z}} (j - \theta)(2R)^{-j} w_c(R^{\theta - j})}{S_c(\theta)} \tag{2}$$

converge absolutely and locally uniformly. The function D_c is continuous and satisfies $D_c(\theta+1) = D_c(\theta)$. Moreover,

$$\boxed{\mathbb{E}[K \mid L = n] = \log_R n + D_c(\log_R n) + o(1).}$$

An exact characterization of w_c is given in (9)*. No claim that D_c is nonconstant is needed.*

2.1 From inverse pruning to immigration

The established inverse-pruning construction gives

$$X_{k+1} = 2X_k + V_k, \qquad V_k \mid (X_k, Y_k) \sim \mathrm{Poisson}(\gamma(c-1)Y_k),$$

where, conditional on $X_k = m$, total edge length satisfies $Y_k \sim \mathrm{Gamma}(2m - 1, \mathrm{rate}\ \gamma)$. Consequently,

$$\mathbb{E}[z^{V_k} \mid X_k = m] = [c - (c-1)z]^{-(2m-1)}. \tag{3}$$

Thus V_k conditional on $X_k = m$ is a sum of $2m-1$ independent copies of G, where

$$\mathbb{P}(G = r) = c^{-1}(1-c^{-1})^r, \qquad r = 0, 1, \dots.$$

Put $\xi = 2 + 2G$. Its mean is R, its minimum is two, and its generating function is

$$f(s) = \mathbb{E}[s^\xi] = \frac{s^2}{c - (c-1)s^2}. \tag{4}$$

Define a separate Markov chain by

$$Z_1 = 1, \qquad Z_{k+1} = 1 + \sum_{i=1}^{Z_k} \xi_{k,i}, \tag{5}$$

with independent offspring variables. Then, by induction on the one-dimensional marginal distributions,

$$X_k \overset{d}{=} \frac{Z_k + 1}{2}. \tag{6}$$

In particular, this step does not require claiming that the original leaf-count sequence is Markov with respect to its inherited tree filtration.

If $F_k(z) = \mathbb{E}[z^{X_k}]$, equivalent exact formulas are

$$F_1(z) = z, \qquad F_{k+1}(z) = [c - (c-1)z] F_k\left(\frac{z^2}{[c-(c-1)z]^2}\right), \tag{7}$$

$$\mathbb{E}[s^{Z_k}] = \prod_{r=0}^{k-1} f^{\circ r}(s), \qquad \mathbb{E}X_k = \frac{R^k + R - 2}{2(R-1)}. \tag{8}$$

Here $f^{\circ 0}(s) = s$. These formulas also fix the order and size normalizations.

3 An elementary uniform local limit theorem

We prove the local result needed for conditioning on an exact number of leaves. Weak convergence alone would not suffice.

Let H_r be a Galton–Watson population with $H_0 = 1$ and offspring generating function (4). Set $U_r = H_r / R^r$.

Lemma 3.1 (Moments and the immigration limit)**.** *For every integer $m \geq 1$, $\sup_r \mathbb{E}U_r^m < \infty$. The martingale U_r converges in every finite L^p to a nonnegative variable U of mean one. If $(U_i)_{i \geq 1}$ are independent copies of U, then*

$$\frac{X_k}{R^k} \implies W_c := \frac{1}{2}\sum_{i=1}^{\infty} R^{-i} U_i, \qquad \mathbb{E}W_c = \frac{1}{2(R-1)}. \tag{9}$$

All integer moments of X_k / R^k are bounded uniformly in k.

Proof. The offspring distribution has every moment. Conditional expansion of a sum of n independent offspring variables gives

$$\mathbb{E}[H_{r+1}^m \mid H_r = n] = R^m n^m + O_m(1 + n^{m-1}).$$

The coefficients implicit in the bound depend only on m, c. Induction on m, followed by division by $R^{m(r+1)}$, bounds the successive increments of $\mathbb{E}U_r^m$ above by a constant times R^{-r}. Their sum is finite. The martingale convergence theorem, applied with a larger integer moment when necessary, gives convergence in each finite L^p and $\mathbb{E}U = 1$.

For independent families indexed by their immigration time,

$$\frac{Z_k}{R^k} \stackrel{d}{=} \sum_{i=1}^{k} R^{-i} U_{k-i}^{(i)}.$$

Couple each family along its own Galton–Watson martingale. For fixed finitely many i the terms converge in L^p; the remaining tail is bounded in L^p by a constant times $\sum_{i>M} R^{-i}$. Hence the sum converges in L^p to $\sum_{i\geq 1} R^{-i} U_i$. Formula (6) proves the assertion and the moment bounds. □

Lemma 3.2 (Uniform characteristic-function decay)**.** *Put $\beta = \log_R 2 \in (0,1)$. There are constants $C, b > 0$, depending on c, such that for every $r \geq 1$,*

$$\left|\mathbb{E}e^{itU_r}\right| \leq Ce^{-b|t|^\beta} \quad \textit{whenever } |t| \leq \pi R^r/2. \tag{10}$$

The limit U has a bounded smooth density.

Proof. Write $\sigma^2 = \mathrm{Var}(\xi) > 0$. The variance recursion gives

$$\mathrm{Var}(U_r) = \frac{\sigma^2}{R(R-1)}(1 - R^{-r}),$$

so these variances are bounded away from zero for $r \geq 1$. If U_r' is an independent copy, the third-moment bound and Taylor's inequality for cosine give, uniformly in $r \geq 1$,

$$|\mathbb{E}e^{itU_r}|^2 = \mathbb{E}\cos(t(U_r - U_r')) = 1 - \mathrm{Var}(U_r)t^2 + O(|t|^3).$$

Thus for some $\delta, \kappa > 0$, $|\mathbb{E}e^{itU_r}| \leq \exp(-\kappa t^2)$ for $|t| \leq \delta$.

For $|z| \leq 1$,

$$|f(z)| \leq |z|^2, \tag{11}$$

and $|f(e^{i\theta})| < 1$ for $0 < |\theta| \leq \pi/2$. Let $t = R^r\theta$. For $|\theta| < \delta/R$ and $|t| \geq \delta$, choose the first integer $j \geq 1$ such that $R^j|\theta| \geq \delta/R$. Then $j \leq r$ and $R^j|\theta| \leq \delta$, so

$$|f^{\circ j}(e^{i\theta})| \leq \exp(-\kappa\delta^2/R^2) =: \alpha < 1.$$

Applying (11) another $r - j$ times gives a bound $\alpha^{2^{r-j}} \leq \exp(-b_1|t|^\beta)$. For $\delta/R \leq |\theta| \leq \pi/2$, compactness bounds $|f(e^{i\theta})|$ by a constant less than one, and the same repeated-squaring argument gives the desired bound. Bounded t is absorbed in C.

Pass to the martingale limit for each fixed t. The limiting characteristic function is bounded by $Ce^{-b|t|^\beta}$; its product with every power of $|t|$ is integrable. Fourier inversion therefore supplies a bounded C^∞ density. □

Lemma 3.3 (Strict positivity)**.** *The density of U is strictly positive at every $x > 0$. The same is true of the density w_c of W_c.*

Proof. Let v be the continuous density of U. It is nonnegative, has total integral one, and is supported on $[0, \infty)$. It is positive on some interval (a, b) with $0 < a < b$. The branching identity is

$$U \stackrel{d}{=} R^{-1} \sum_{i=1}^{\xi} U_i.$$

For every $m \geq 1$, the event $\xi = 2m$ has positive probability. Its convolution term shows that v is positive throughout $(2ma/R, 2mb/R)$. For all sufficiently large m these intervals overlap and cover a half-line (A, ∞). The event $\xi = 2$ then gives positivity on $(2A/R, \infty)$. Iteration, using $2/R < 1$, gives positivity for every $x > 0$. These pointwise inferences are valid because

the density and each individual convolution term are continuous; an almost-everywhere lower bound between them is therefore pointwise.

For the immigration limit, write $2W_c = R^{-1}U_1 + V$ with $V = \sum_{i\geq 2} R^{-i}U_i$ independent of U_1. For every $x > 0$, $\mathbb{P}(V < x) > 0$: make finitely many summands small, using positivity of the density of U near zero, and then use the expectation bound for the remaining independent tail. Consequently

$$w_c(x) = 2R\,\mathbb{E}\Big[v(R(2x - V))\,\mathbf{1}_{\{V<2x\}}\Big] > 0.$$

Boundedness and continuity follow either from this convolution or from the characteristic-function argument below. □

Proposition 3.4 (Local limit with weighted local bounds)**.** *The density w_c in Theorem 2.1 satisfies* (with $w_c(x) = 0$ for $x \leq 0$)

$$\sup_{n\in\mathbb{Z}} \Big| R^k \mathbb{P}(X_k = n) - w_c(n/R^k) \Big| \longrightarrow 0. \tag{12}$$

For each integer $m \geq 0$ there is $C_m < \infty$ such that

$$R^k \mathbb{P}(X_k = n) \leq C_m (1 + n/R^k)^{-m} \qquad (k \geq 1,\ n \geq 1). \tag{13}$$

The analogous bound holds for $w_c(x)$, $x \geq 0$.

Proof. Let $\Phi_k(t) = \mathbb{E}\exp(itX_k/R^k)$. The independent-family representation writes Φ_k as a deterministic phase times the characteristic functions of the scaled family variables $\frac{1}{2}R^{-i}U^{(i)}_{k-i}$. The oldest family and Lemma 3.2 imply

$$|\Phi_k(t)| \leq C \exp(-b'|t|^\beta), \quad |t| \leq \pi R^k, \quad k \geq 2. \tag{14}$$

The limiting characteristic function is that of W_c. Extend Φ_k by zero outside this interval. Dominated convergence gives convergence in $L^1(\mathbb{R})$ to the limiting characteristic function. Since X_k is integer-valued, lattice Fourier inversion gives

$$R^k \mathbb{P}(X_k = n) = \frac{1}{2\pi} \int_{-\pi R^k}^{\pi R^k} \Phi_k(t) e^{-itn/R^k}\, dt.$$

The L^1 convergence proves (12), uniformly in n.

For completeness, the following derivative estimate supplies the global local tails required for conditioning. Fix m and suppose $k \geq m + 2$. Separate the oldest $m + 1$ independent family factors from the rest. Their generations are $k - 1, \ldots, k - m - 1$, all at least one. Differentiate the product m times. In every term at least one of these $m + 1$ factors is undifferentiated. That factor retains the exponential bound of Lemma 3.2 throughout $[-\pi R^k, \pi R^k]$. Every differentiated factor is bounded by a moment of its scaled family variable; the factor containing the remaining families has the same property by Lemma 3.1. There are only finitely many product-rule terms for fixed m. Thus

$$|\Phi_k^{(m)}(t)| \leq C'_m \exp(-b_m|t|^\beta) \quad (|t| \leq \pi R^k), \tag{15}$$

with constants independent of k. Indeed, for retained family $i \leq m + 1$, its Fourier argument is $t/(2R^i)$; the permitted window is $|t|/(2R^i) \leq \pi R^{k-i}/2$, exactly $|t| \leq \pi R^k$. Its decay constant may depend on fixed i, but not on k. This checks the full inversion window for every possible undifferentiated factor.

Integrate the Fourier inversion formula by parts m times. All boundary terms cancel: for each derivative order the boundary expressions involve $\mathbb{E}[(X_k/R^k)^j e^{\pm i\pi X_k}]e^{\mp i\pi n}$, whose two signs give the same value since X_k, n are integers. It follows that

$$(n/R^k)^m R^k \mathbb{P}(X_k = n) \leq \frac{1}{2\pi} \int_{-\pi R^k}^{\pi R^k} |\Phi_k^{(m)}(t)|\, dt \leq C''_m.$$

Combine this with the zeroth-order bound. The finitely many smaller k have generating functions analytic in a neighborhood of 1, by (7), and hence exponentially decreasing point probabilities. Enlarging the constant covers them. This proves (13). Passing to the local limit proves the same bound for w_c. □

4 Conditioning on the exact leaf count

Proof of Theorem 2.1. Put $m_n = \lfloor \log_R n \rfloor$ and $\theta_n = \log_R n - m_n \in [0, 1)$. For $k = m_n + j \geq 1$,

$$\begin{aligned}(2R)^{m_n}\mathbb{P}(K = m_n + j, L = n) &= (2R)^{-j}\, R^{m_n+j}\mathbb{P}(X_{m_n+j} = n)\\ &\longrightarrow (2R)^{-j} w_c(R^{\theta_n - j}),\end{aligned}$$

where the displayed asymptotic means that the difference tends to zero for each fixed j, uniformly in the phase occurring on the right.

The local bound in Proposition 3.4, with any integer $M > 1 + \log_R 2$, supplies a summable majorant independent of n and θ_n. For $j \geq 0$ use $C(2R)^{-j}$. For $j < 0$ use

$$C(2R)^{-j}R^{Mj} = C\left(\frac{R^M}{2R}\right)^j.$$

These bounds remain summable after multiplication by any fixed power of $|j|$. The missing terms with $m_n + j < 1$ are covered by the same vanishing tail. Dominated summation proves, uniformly in the phase,

$$\begin{aligned}(2R)^{m_n}\mathbb{P}(L = n) &= S_c(\theta_n) + o(1),\\ \mathbb{E}[K - m_n \mid L = n] &= \frac{\sum_j j(2R)^{-j} w_c(R^{\theta_n - j})}{S_c(\theta_n)} + o(1).\end{aligned}$$

Strict positivity and continuity of w_c give $\inf_{0\leq\theta\leq 1} S_c(\theta) > 0$, for example using just its $j = 0$ term. This justifies division. The same summable bounds establish continuity of the defining series. Replacing j by $j+1$ shows that both numerator and denominator in (2) acquire the same factor $(2R)^{-1}$ when θ is replaced by $\theta + 1$. Hence D_c is period one. Finally, $m_n = \log_R n - \theta_n$ gives the theorem. □

Corollary 4.1. *The conditional distribution of $K - m_n$ is asymptotically the phase-dependent probability mass function*

$$p_\theta(j) = \frac{(2R)^{-j} w_c(R^{\theta - j})}{S_c(\theta)}, \qquad j \in \mathbb{Z}.$$

The approximation holds in total variation along the actual phases θ_n, and for every fixed absolute moment. Also,

$$\mathbb{P}(L = n) = n^{-(1+\log_R 2)}\big[P_c(\log_R n) + o(1)\big], \quad P_c(\theta) = (2R)^\theta S_c(\theta),$$

where P_c is positive, continuous and period one.

II. Strong geometric Horton asymptotics

5 Kingman: deterministic densities and the ODE hierarchy

For an N-leaf Kingman coalescent tree, let $N_j^{(N)}$ be the number of order-j branches. The established finite-order result in [5] identifies constants

$$\frac{N_j^{(N)}}{N} \xrightarrow[N\to\infty]{\mathbb{P}} \mathcal{N}_j, \qquad \mathcal{N}_j = \frac{1}{2}\int_0^\infty g_j(t)^2\, dt,$$

where

$$g_1(t) = \frac{2}{t+2}, \qquad g'_{j+1} = \frac{1}{2}g_j^2 - g_j g_{j+1}, \qquad g_{j+1}(0) = 0.$$

Define $h_0 = 0$ and, for $k \geq 1$,

$$h'_k = 2h_{k-1}h_k - h_{k-1}^2, \qquad h_k(0) = 1. \tag{16}$$

In particular, $h_1 = 1$ and $h_2 = (1 + e^{2x})/2$. The transformation

$$h_k(x) = \frac{1}{1-x} - \frac{g_{k+1}(2x/(1-x))}{(1-x)^2}$$

gives, with the indexing made explicit,

$$A_k := \int_0^1 [1 - (1-x)h_k(x)]^2 \, dx = \mathcal{N}_{k+1}, \qquad A_0 = \mathcal{N}_1 = 1. \tag{17}$$

Every finite h_k is smooth and finite on $[0, 1]$, by induction through the linear ODE. Let $a_k = h_k(1)$ for $k \geq 1$.

Theorem 5.1. *There exist $\rho \in [2, 4]$ and $H \in (0, \infty)$ such that*

$$\boxed{\frac{\mathcal{N}_j}{\mathcal{N}_{j+1}} \longrightarrow \rho, \qquad \mathcal{N}_j \rho^j \longrightarrow H.}$$

More explicitly,

$$\rho = \lim_{k\to\infty} \frac{a_{k+1}}{a_k}, \qquad H = \frac{\rho B_*}{C_*},$$

where B_* and C_* are positive monotone limits constructed below. In particular $\rho/3 \leq H \leq \rho^2$.

5.1 Basic comparison facts

Subtracting consecutive equations gives

$$(h_{k+1} - h_k)' = 2h_k(h_{k+1} - h_k) + (h_k - h_{k-1})^2.$$

Hence $0 \leq h_{k-1} \leq h_k$, with strict order for $x > 0$ once applicable. Therefore

$$0 \leq h'_k = h_k^2 - (h_k - h_{k-1})^2 \leq h_k^2, \qquad h_k(x) < \frac{1}{1-x} \quad (0 < x < 1). \tag{18}$$

For later comparisons, if two ordered hierarchies have consecutive entries (p, u) and (q, v), their difference obeys the identity

$$(2pu - p^2) - (2qv - q^2) = (p+q)(u-v) + (p-q)(u+v-p-q). \tag{19}$$

When $p \geq q$, $u \geq p$, $v \geq q$ and the initial difference $u - v$ is nonnegative, the integrating-factor formula implies $u - v \geq 0$ throughout the interval. The forcing sign uses $u + v - p - q = (u - p) + (v - q) \geq 0$ in the ordered region; it does not assume the conclusion $u \geq v$ in advance.

5.2 Endpoint rescaling and its monotonicity

For $\gamma \in (0,1)$ set

$$V_{k,\gamma}(x) = \begin{cases} (1-x)^{-1}, & 0 \le x \le 1-\gamma, \\ \gamma^{-1} h_k((x-1+\gamma)/\gamma), & 1-\gamma \le x \le 1. \end{cases}$$

This is the comparison family used in [5, 6]; a proof of its needed properties follows to make the analytic argument explicit.

Lemma 5.2. *For $\gamma_k = a_k/a_{k+1}$,*

$$h_{k+1} \le V_{k,\gamma_k}, \qquad \gamma_{k+1} \ge \gamma_k.$$

Consequently a_{k+1}/a_k decreases to a finite limit $\rho \ge 1$.

Proof. The function $(1-x)h_k(x)$ is nonincreasing by (18). For $y = (x-1+\gamma)/\gamma \le x$ and $x \ge 1-\gamma$, this implies $\gamma^{-1}h_k(y) \ge h_k(x)$. Thus $V_{k,\gamma} \ge h_k$, and $V_{k+1,\gamma} \ge V_{k,\gamma}$. Both hierarchies satisfy the same ODE relation, with the V relation understood piecewise at $1-\gamma$.

For fixed γ, the function $D_k = V_{k,\gamma} - h_{k+1}$ changes sign at most once, from positive to negative, on $[1-\gamma, 1]$. At $k=1$, $V_{1,\gamma}$ is constant on that interval and h_2 is increasing. Inductively,

$$\frac{d}{dx}\left[D_{k+1} e^{-2\int_{1-\gamma}^{x} h_{k+1}}\right] = (2V_{k+1,\gamma} - V_{k,\gamma} - h_{k+1}) D_k e^{-2\int_{1-\gamma}^{x} h_{k+1}}.$$

The first factor is nonnegative: $V_{k+1,\gamma} \ge V_{k,\gamma}$ and $V_{k+1,\gamma} \ge h_{k+1}$. The integrating-factor expression starts strictly positive and first increases, then can only decrease. It can therefore cross zero at most once, in the stated direction; once negative, it cannot return to zero. Taking $\gamma = \gamma_k$ makes $D_k(1) = 0$, so D_k cannot have been negative before the endpoint. On the remaining interval the inequality follows from (18).

Now compare V_{k+1,γ_k} with h_{k+2}. The preceding levels satisfy $V_{k,\gamma_k} \ge h_{k+1}$, both hierarchies are ordered, and both next levels start at one. Formula (19) gives $V_{k+1,\gamma_k} \ge h_{k+2}$. At the endpoint this reads $a_{k+1}/\gamma_k \ge a_{k+2}$, or $\gamma_{k+1} \ge \gamma_k$. Positivity and $a_{k+1} \ge a_k$ finish the proof. □

6 Monotone normalized profiles

For $u \ge 0$ define the continuous extension

$$q_k(u) = \begin{cases} a_k^{-1} h_k(1-u/a_k), & 0 \le u \le a_k, \\ u^{-1}, & u \ge a_k. \end{cases} \tag{20}$$

At $u = 0$ the first branch is used. Lemma 5.2 gives

$$q_{k+1}(u) \le q_k(u) \quad \text{for every } u \ge 0. \tag{21}$$

Indeed, for $u \le a_k$ this is exactly $h_{k+1} \le V_{k,\gamma_k}$ after substituting $x = 1 - u/a_{k+1}$. For $a_k \le u \le a_{k+1}$ it follows from $h_{k+1}(x) \le (1-x)^{-1}$, and beyond a_{k+1} both extensions equal $1/u$.

Furthermore, (18) and $q_k(0) = 1$ give

$$\frac{1}{1+u} \le q_k(u) \le \min\{1, u^{-1}\}. \tag{22}$$

For the lower bound, on the first interval $-q_k^2 \le q_k' \le 0$, so $(1/q_k)' \le 1$; the extension also satisfies the bound.

Put $e_k(u) = 1 - uq_k(u)$, with $e_k(0) = 1$. Then e_k vanishes for $u \ge a_k$, increases pointwise with k, and

$$0 \le e_k(u) \le (1+u)^{-1}.$$

The change of variables $u = a_k(1-x)$ yields

$$B_k := a_k A_k = \int_0^\infty e_k(u)^2\,du \;\uparrow\; B_* \in [1/3, 1]. \tag{23}$$

The upper bound is the integral of $(1+u)^{-2}$; the lower bound follows already from $B_1 = \int_0^1 (1-u)^2\,du = 1/3$.

This step alone gives the ratio law, since

$$\frac{A_k}{A_{k+1}} = \frac{a_{k+1}}{a_k}\frac{B_k}{B_{k+1}} \longrightarrow \rho.$$

It does not, by itself, prove positivity of the strong-law constant. That is the purpose of the next comparison.

7 A fractional-linear comparison and the positive strong limit

Lemma 7.1 (Supermultiplicativity)**.** *For all integers $\ell, j \geq 1$,*

$$a_{\ell+j} + 1 \geq (a_\ell + 1)(a_j + 1). \tag{24}$$

Proof. For $0 < b < 1$, put $a = 1 - b$ and define

$$(\mathcal{M}_b h_j)(x) = \frac{b}{1-bx} + \frac{a}{(1-bx)^2} h_j\!\left(\frac{ax}{1-bx}\right), \qquad 0 \leq x \leq 1. \tag{25}$$

The transformed argument runs from zero to one. Direct differentiation shows that the transformed hierarchy satisfies the same ODE (16). Its entries of index $j \geq 1$ start at $b + a = 1$, and its index-zero entry is $b/(1-bx)$. The hierarchy remains ordered, since its scale factor is positive.

Fix ℓ. Integrating $h_\ell'/h_\ell^2 \leq 1$ from x to 1 gives

$$h_\ell(x) \geq \frac{a_\ell}{1 + a_\ell(1-x)}.$$

Choose $b = a_\ell/(1+a_\ell)$, so this lower bound is exactly the index-zero entry $b/(1-bx)$ of the transformed hierarchy. Compare the shifted hierarchy $h_{\ell+j}$ with $\mathcal{M}_b h_j$ using (19), inductively in j. Their next-level initial values agree, and their level-zero entries are ordered, hence

$$h_{\ell+j}(x) \geq (\mathcal{M}_b h_j)(x).$$

At $x = 1$, the right side is $a_\ell + (1 + a_\ell)a_j$. Adding one gives (24). □

By Lemma 5.2, $a_k^{1/k} \to \rho$. Supermultiplicativity and $a_1 + 1 = 2$ imply $a_k + 1 \geq 2^k$, hence $\rho \geq 2$. Iterating (24) at any fixed index k and taking roots gives

$$a_k + 1 \leq \rho^k \qquad (k \geq 1). \tag{26}$$

Since every endpoint ratio is at least its decreasing limit ρ,

$$C_k := a_k \rho^{-k} \;\uparrow\; C_*, \qquad \frac{1}{\rho} \leq C_* \leq 1. \tag{27}$$

This bounded, positive normalization excludes the otherwise possible zero strong limit.

Proof of Theorem 5.1. Using (17), (23) and (27),

$$\mathcal{N}_j \rho^j = A_{j-1}\rho^j = \rho\,\frac{B_{j-1}}{C_{j-1}} \longrightarrow H := \frac{\rho B_*}{C_*} \in (0,\infty).$$

The ratio law follows either from this positive limit or from the direct computation in the preceding section. The bounds on B_* and C_* give $\rho/3 \le H \le \rho^2$.

For completeness, to obtain $\rho \le 4$, set $r_k(x) = 1-(1-x)h_k(x)$. Equation (16) gives

$$[(1-x)r_{k+1}(x)]' = r_k(x)^2 - 2r_k(x)r_{k+1}(x).$$

Both boundary values on the left are zero. Thus $A_k = 2\int_0^1 r_k r_{k+1}\,dx$, and Cauchy–Schwarz yields $A_k \le 4A_{k+1}$. Passing to the ratio limit gives $\rho \le 4$. The same identity and $r_k \ge r_{k+1} \ge 0$ give the familiar lower bound $A_k \ge 2A_{k+1}$. □

Remark 7.2 (Exact characterizations versus decimal estimates). The proof supplies exact definitions

$$\rho = \inf_{k\ge1}\frac{a_{k+1}}{a_k} = \sup_{k\ge1}(a_k+1)^{1/k}, \qquad H = \rho\,\frac{\int_0^\infty (1-uq(u))^2\,du}{\sup_k a_k\rho^{-k}},$$

where $q(u) = \inf_k q_k(u)$. These characterize the constants without requiring a closed form. In particular, exact endpoint evaluations would give the rigorous enclosure $(a_k+1)^{1/k} \le \rho \le a_{k+1}/a_k$.

8 Checks, limitations and reproducibility

The proofs above are analytic. The supplied scripts are independent diagnostics of algebra, indexing and finite computations, not a formal proof certificate or an independent referee report.

Exact checks. `verify_exact.py` verifies the fractional-linear ODE identity and the cooperative comparison identity symbolically. It computes finite leaf-count probabilities with rational arithmetic for $c = 5/4, 3/2, 2, 7/2$ through $n = 64$, checks positivity and minimum-size support, and verifies at $c = 2$ that the order mixture equals

$$\mathbb{P}(L=n) = \frac{1}{2^{2n-1}}\frac{1}{n}\binom{2n-2}{n-1} \qquad (1 \le n \le 64).$$

This last identity is a finite check against the critical binary Galton–Watson/Catalan case, not an input to the general-c proof.

Numerical checks. `verify_numeric.py` integrates a rescaled form of the hierarchy. It uses $s = -\log(1-x)$ and $z_k(s) = \log h_k(1-e^{-s})$, avoiding direct resolution of exponentially narrow endpoint layers. Finite-order values and floating-point diagnostics give

$$\rho \approx 3.04382786, \qquad B_* \approx 0.52476436, \qquad C_* \approx 0.51627578, \qquad H \approx 3.09387432.$$

These decimals are not interval-certified. The revised script uses `expm1` for the small residual, a central integration window, and retained quadrature errors and warnings. The profile envelope bounds omitted tails for the window $[-25,25]$ by $2/(1+e^{25}) < 2.78\times10^{-11}$; this is not an ODE or roundoff error bound. The original 60-order diagnostic drifted by about 0.245% in H while still passing finite inequalities. That evidence is retained in the review audit. The revised `verify_stability.py` separately compares 40- and 60-order runs with different step sizes. Finite inequalities, quadrature diagnostics and empirical stability are distinct checks. All checks remain active under optimized Python. Neither apparent convergence rates nor a finite grid is used to justify an infinite limit.

Status. This is a candidate complete proof of the two stated mathematical questions under the source conventions. A supplied review and internal analytical and computational checks support it; these are not external peer review, independent reproduction, or proof-assistant verification. The novelty comparison is targeted rather than exhaustive. Nothing here proves an unrestricted joint j, N limit for finite Kingman trees or empirically establishes that real river networks follow either model.

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